

Refer to the **insert** for the list of pseudocode functions and operators.

- 1 A user requires a program that implements complex data encryption to a recognised international standard. A programmer has started the design of the program.

- (a) The programmer decides to use a library routine to provide part of the solution. One reason for this is that library routines may perform functions that the programmer is unable to program themselves.

State **three** other benefits of using library routines in the development of the program.

- 1
- 2
- 3
- [3]

- (b) The programmer has identified a different part of the algorithm that would be appropriate to implement as a subroutine (a procedure or a function); no suitable library routine exists for this part.

- (i) State **two** reasons why the programmer may decide to use a subroutine.

- 1
- 2
- [2]

- (ii) A function header in pseudocode is defined as:

FUNCTION Pass2 (Count : INTEGER) RETURNS BOOLEAN

Complete the table by describing the terms used in the function header.

The first row has been completed.

Term	Description
Pass2	the name of the function
Count	
BOOLEAN	

[2]

